

开环模型

gain crossover

相角裕度

phase crossover

增益裕度

interpretation

$$G(s) = \frac{100}{s(s+2)(s+10)}$$

$$\omega_c \approx 2.8 \text{ from } |G(j\omega_c)| = 1$$

$$\gamma = 180^\circ + \angle G(j\omega_c) = 19.9^\circ$$

$$\omega_g^2 = 20 \Rightarrow \omega_g = 4.47$$

$$h = \frac{1}{|G(j\omega_g)|} = 2.4 \text{ (7.6 dB)}$$

系统稳定，但相位储备偏紧，不宜再继续降低